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Studierichting: Informatica
Oefeningenreeks 5K nr. 1.5

$$r = \sin^3\left(\frac{\theta}{3}\right) \text{ in het interval } [0, 3\pi]$$

$$r^2 = \sin^6\left(\frac{\theta}{3}\right)$$

$$(r')^2 = \sin^4\left(\frac{\theta}{3}\right) \cdot \cos^2\left(\frac{\theta}{3}\right)$$

$$\Rightarrow \int_{\alpha}^{\beta} \sqrt{r^2 + (r')^2}d\theta = \int_{0}^{3\pi} \sqrt{\sin^6\left(\frac{\theta}{3}\right) + \sin^4\left(\frac{\theta}{3}\right) \cdot \cos^2\left(\frac{\theta}{3}\right)}d\theta$$

$$= \int_{0}^{3\pi} \sqrt{\sin^4\left(\frac{\theta}{3}\right)\left(\sin^2\left(\frac{\theta}{3}\right) + \cos^2\left(\frac{\theta}{3}\right)\right)}d\theta$$

$$= \int_{0}^{3\pi} \sqrt{\sin^4\left(\frac{\theta}{3}\right)}d\theta$$

$$= \int_{0}^{3\pi} \sin^2\left(\frac{\theta}{3}\right) d\theta$$

$$= \frac{1}{2}\int_{0}^{3\pi} \left(1 + \cos\left(\frac{2\theta}{3}\right)\right) d\theta$$

$$= \left[\frac{\theta}{2}\right]_0^{3\pi} + \frac{3}{4}\left[\sin\left(\frac{2\theta}{3}\right)\right]_0^{3\pi}$$

$$= \frac{3\pi}{2}$$

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References